

Current limiting algorithms and transient stability analysis of grid-forming VSCs

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ABSTRACT

This paper consists first in comparing the well-known virtual impedance and current saturation algorithms in case of a three-phase short circuit. During the fault, the virtual impedance has an indirect action on the AC current since it depends mainly on the AC voltage controllers dynamics, whereas, the current saturation algorithm has a direct action on the AC current. Thus, a first study consists in comparing the dynamic behavior of the AC current for both methods separately. Once the fault is cleared, the grid-forming converter should be able to resynchronize to the main grid. In this paper, the impact of each current limiting algorithm on the transient stability is highlighted. Based on this comparative approach between the virtual impedance and the current saturation algorithms, a hybrid current limiting control is proposed to benefit the advantages of each algorithm in terms of current limitation and transient stability. To show the effectiveness of the proposed control, time-domain simulations and experimental test cases have been performed.

1. Introduction

Grid-forming Voltage Source Converter (VSC) based on droop control [1] has been widely studied in Microgrid and uninterruptible power supply for several types of applications [2,3]. These studies have been recently extended to power transmission systems with new challenges [4,5]. Grid-forming VSCs are able to generate an instantaneous AC voltage similarly to synchronous generators (SGs), and operate in both grid-connected and autonomous modes. However, unlike grid-following converters based on phase-locked loop (PLL) that behave as current sources, grid-forming converters are very sensitive to external disturbances. Indeed, due to the voltage source behavior of a grid forming converter, the overcurrent protection deserves a specific attention [6,7]. Compared to SG that can support up to seven times over its rated current, power converters can only cope with few percent of overcurrent 20–40% [8]. Trying to mimic synchronous machine would require a very large oversizing of semiconductor components and induces large additional costs. Therefore, power converters have to be protected against extreme events such as short circuits but also other events which may induce smaller overcurrent such as phase shift only based on control laws.

Two main strategies have been proposed for grid-forming converter in order to limit the current during transients. One solution is to limit the current with a current saturation algorithm (CSA). This control

strategy has been implemented in different ways e.g.; [6,9,10]. It has a direct action on the current. Therefore, it ensures an accurate current limitation when the current crosses its maximum allowable magnitude. The second solution is based on the Virtual Impedance (VI) [7,11–13]. It has been implemented to emulate the effect of physical impedance when the current exceeds its nominal value. This method has shown its effectiveness to limit the current transients in case of a phase shift and a heavy load connection [7]. This control strategy consists to modify the AC voltage reference in order to limits the current. Therefore, its reaction against the fault depends on the AC current dynamics as well as on the dynamics of the AC voltage regulators.

Once the current is limited, the post-fault synchronization is of highest importance. Indeed, an unstable behavior when the overcurrent protection is disabled would lead to an unavoidable trip from the main grid for safety purposes [14]. However, in future power systems all different power sources have to cope with various events and guaranty the stability. In this context, the transient stability of the system is defined as the ability of the power converter to recover the synchronism after fault clearing. The choice of the current limiting algorithm has a large influence on the transient stability. In [15,16], the transient instability phenomenon using the CSA while keeping the droop control has been studied. To avoid this instability issue, some papers proposed to switch from voltage control based on the droop control to current control based on a phase-locked loop [1,17]. These methods require a

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complex algorithm for fault detection, triggering conditions. Moreover, as specified by [1], fault recovery could only be achieved after a considerable retuning of the voltage controller and the power controller in order to avoid AC voltage collapse. In [7], the authors showed through time-domain simulations the ability of the VI to ensure a better transient stability than the CSA in case of a heavy load connection, however, this result has not been theoretically investigated.

In this paper, the focus is made on the droop control based-VSC with a negligible inertial effect when the system is subjected to a three-phase bolted fault. The reactive droop control is null, and thus, the AC voltage is considered constant throughout the fault. The contributions of the present paper consist to:

- Compare the CSA and VI when the current is limited.
- Study the impact of the CSA and the VI on the transient stability of the system.
- Propose a hybrid current limiting control (HCLC) to ensure an accurate current limitation thanks to the CSA and a better transient stability thanks to the VI.

To show the effectiveness of the proposed method, experimentations are performed in “EPMLab platform [18]”.

The reminder of this paper is organized as follow. Section 2 first describes the studied system and its control structure. Section 3 recalls the operation principle of the virtual impedance and the CSA for current limitation and shows their impact on the current dynamic during the fault. Section 4 presents the synchronization process of the grid-forming based on the droop control embedding the current limiting algorithms (i.e., VI and CSA). Section 5 presents a hybrid current limiting control and its experimental validation.

2. Grid-forming VSC based on droop control

2.1. Description of the system

The studied system is depicted in Fig. 1. It consists of a three-phase 2-Level VSC supplied by a DC voltage source. On the AC side, the VSC is connected to an AC grid through an $L_f C_f L_c$ filter. The AC grid is represented by an ideal AC voltage source in series with its equivalent impedance $L_g R_g$.

v_m represents the modulated voltage by the converter, i_s the output current through the filter L_f , e_g the AC voltage across the capacitor C_f , i_g the grid current through the transformer inductance L_c , v_{pcc} the AC voltage at the point of common coupling (PCC), v_g the grid voltage and u_{dc} the DC voltage.

2.2. Grid-forming control structure

Fig. 2 presents the grid-forming cascaded control structure. The control contains an inner cascaded AC voltage and current loops represented in the synchronous rotating frame (SRF) [19]. The inner control is ensured by two cascaded proportional-integral controllers (PI), feedforward decoupling terms and compensations. The AC voltage

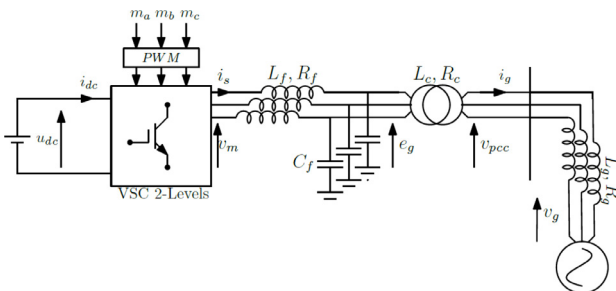


Fig. 1. Power electronic converter connected to an AC system via an LCL filter.

loop determines the current references. These latter are sent to the current control that generates the modulated voltage to the linearization stage, and then delivers the modulation signals to the switching stage of the power converter. The control angle θ_{VSC} is provided by a primary control based on the droop control $P - \omega$. The primary control is expressed by the following equation:

$$\omega_{VSC} - \omega_{set} = m_p \frac{\omega_c}{\omega_c + s} (p^* - p_{mes}) \quad (1)$$

ω_b , ω_{seb} , ω_{VSC} and m_p are respectively the nominal frequency in rad/s, the nominal grid frequency in per-unit, the internal frequency and the active droop gains. ω_c denotes the cut-off frequency of the low-pass filters in the active power droop. The role of this low-pass filter is usually twofold:

- Measurement filtering.
- Inertial effect emulation.

In this paper, ω_c is chosen to keep filtering the active power measurement and to have a negligible inertial effect. In addition, CSA and VI have been directly integrated in the control structure (Fig. 2) and can be individually enabled-disabled.

3. Current limiting algorithms

3.1. Recall on the virtual impedance principle

The virtual impedance strategy adopted in this paper is the one proposed in [7].

As illustrated in Fig. 3, the VI operation principle is divided into three main phases:

- Overcurrent detection and activation (i.e.; Activation only when the current exceeds its nominal value $I_n = 1$ p.u).
- Virtual impedance calculation.
- AC voltage drop calculation.

The expressions of X_{VI} and R_{VI} are given in (2a) and (2b).

$$X_{VI} = \begin{cases} k_{p_{RVI}} \sigma_{X/R} \delta I & \text{if } \delta I > 0 \\ 0 & \text{if } \delta I \leq 0 \end{cases} \quad (2a)$$

$$R_{VI} = X_{VI} / \sigma_{X/R} \quad (2b)$$

where $\delta I = I_s - I_n$ represents the overcurrent magnitude. $k_{p_{RVI}}$ and $\sigma_{X/R}$ are defined respectively as the virtual impedance proportional gain and the virtual impedance ratio. The parameter $k_{p_{RVI}}$ is tuned to limit the current magnitude to a suitable level I_{maxVI} during overcurrent in the steady state, while $\sigma_{X/R}$ ensures a good system dynamics during the overcurrent. The tuning method of these parameters is explained in [7]. When the virtual impedance is activated, the new AC voltage references are given by (3) and (4):

$$e_{gd}^* = e_{gd}^* - \delta e_{gd} \quad (3)$$

$$e_{gq}^* = e_{gq}^* - \delta e_{gq} \quad (4)$$

with:

$$\delta e_{gd} = R_{VI} i_{sd} - X_{VI} i_{sq} \quad (5)$$

$$\delta e_{gq} = R_{VI} i_{sq} + X_{VI} i_{sd} \quad (6)$$

e_{gd}^* and e_{gq}^* denote the AC voltage setpoint, which are equal to 1 p.u and 0 p.u, respectively, (i.e., the SRF is aligned with the voltage generated by the VSC).

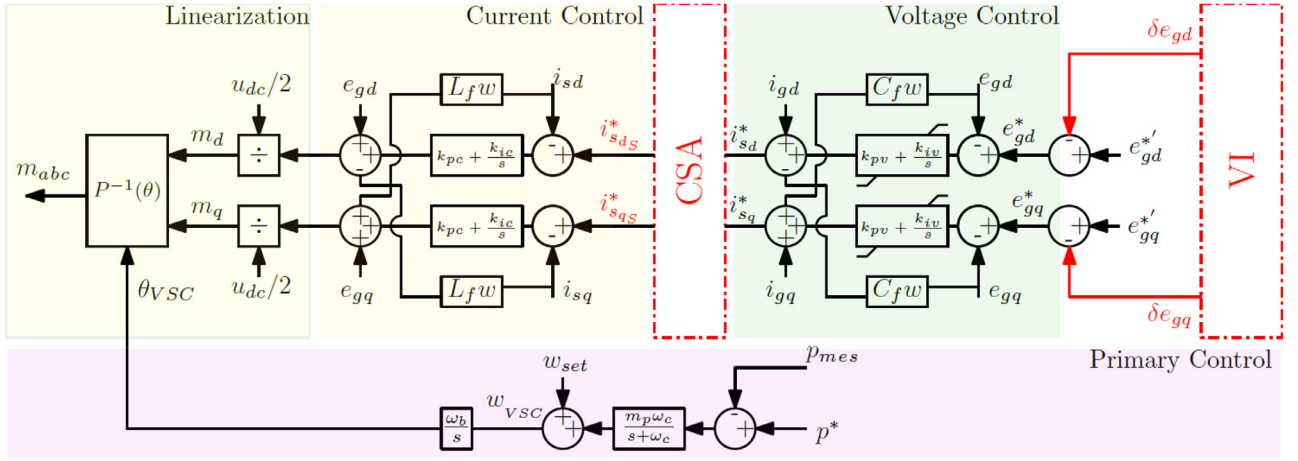


Fig. 2. Grid-forming control structure.

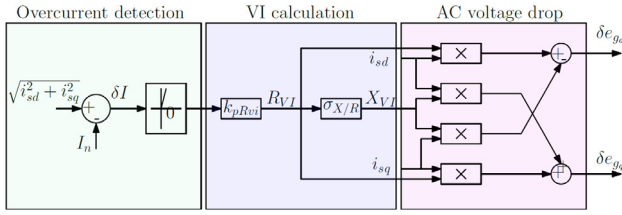


Fig. 3. Virtual impedance principle.

3.2. Recall on the current saturation algorithm

The other way to limit the current in the converter is to apply saturation on the current references. In this paper, the d -axis priority based-CSA is chosen for VSC current limitation [16]. This strategy is described as,

$$\begin{cases} |i_{sd}^*| = \min(I_{\max\text{SAT}}, |i_{sd}^*|) \\ |i_{sq}^*| = \min\left(\sqrt{I_{\max\text{SAT}}^2 - |i_{sd}^*|^2}, |i_{sq}^*|\right) \end{cases} \quad (7)$$

$I_{\max\text{SAT}}$ denotes the maximum allowable current magnitude of the converter. The other variables in (7) are well depicted in Fig. 2.

When the current magnitude is lower than $I_{\max\text{SAT}}$, $i_{sd}^* = i_{sd}^*$ and $i_{sq}^* = i_{sq}^*$, respectively. This operation mode is defined as the unsaturated current mode. To avoid the AC voltage controller windup during the current saturated mode, the integral controller gains k_{iv} are set to zero.

3.3. Comparison between VI and CSA in case of a three phase bolted fault

In order to show the main differences between both strategies in terms of current limitation, time-domain simulations have been carried out using MATLAB/Simulink, where each method is tested separately.

System and control parameters are listed in Table 1,

In Fig. 4, a three-phase bolted fault is applied to the system at the PCC at $t = 1$ s.

When the fault occurs at PCC, v_{pcc} drops to zero, leading to an increase of the current magnitude I_s . During the fault, the VI is supposed to limit the current to the specified value $I_{\max\text{VI}}$. However, the simulation results reveal some oscillations during the first 25 ms. This transient behavior could be dangerous, since it reaches high current peaks of 1.63 p.u. Indeed, the effect of the VI depends highly on the inner voltage loop dynamics, and generally, whatever the AC voltage dynamics is, the response is never fast enough to limit the first current transient. Compared to the VI, the CSA shows its effectiveness to ensure

Table 1
System and control parameters.

Symbol	Value	Symbol	Value
P_n	1 GW	m_p	0.04 p.u.
$\cos\phi$	0.95	k_{ic}	1.19 p.u.
f_n	50 Hz	R_g	0.01 p.u.
U_{ac}	320 kV ph-ph	L_g	0.1 p.u.
$R_f = R_c$	0.005 p.u.	ω_c	62.8 rad/s
$L_f = L_c$	0.15 p.u.	$I_{\max\text{VI}} = I_{\max\text{SAT}}$	1.2 p.u.
C_f	0.066 p.u.	I_n	1 p.u.
k_{pc}	0.73 p.u.	k_{pv}	1 p.u.
$\sigma_{X/R}$	5	k_{iv}	1.16 p.u.
$k_{p_{RVI}}$	0.677 p.u.	f_{sw}	5 kHz

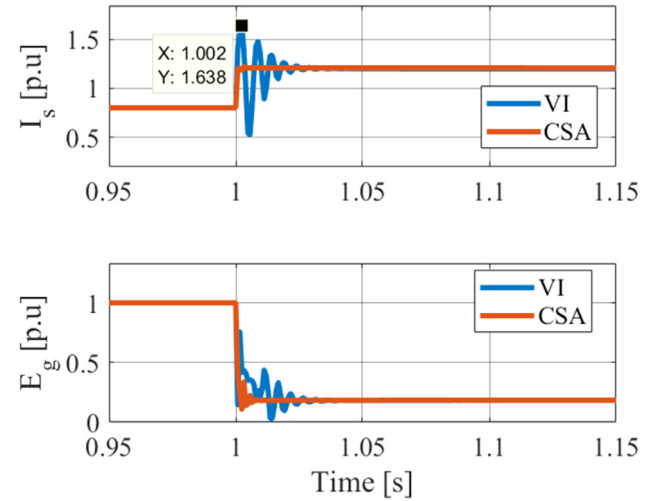


Fig. 4. Three-phase bolted fault at PCC level. Comparison between the CSA and VI.

an accurate current limitation during all the fault duration because of its direct action on the current. From these results, a first conclusion is drawn: CSA is more efficient than VI for current limitation.

4. Post-fault synchronization of a grid-forming converter based on droop control

Once the fault is cleared (i.e., three-phase bolted fault), the power converter has to be able to resynchronize to the AC grid only using local measurements. The aim of this section is to compare the transient

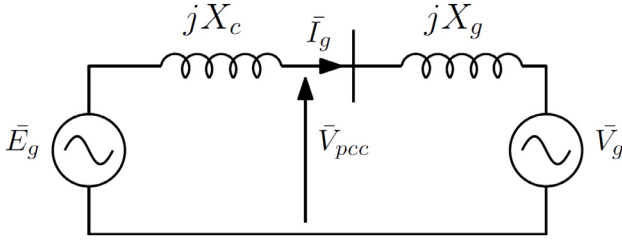


Fig. 5. Simplified quasi-static grid-forming model in unsaturated mode.

stability of the system in case of a bolted fault with VI or CSA. For the sake of simplicity, an assumption is made:

- The $L_f C_f$ filter and the inner control dynamics are ignored.

To check the conclusion drawn from this first analysis, some simulations are proposed with the full system dynamics.

4.1. Transient stability analysis of a VSC based on droop control

A simplified quasi-static model in Fig. 5 is derived from the previous assumptions. In this first approach, the current limiting algorithms are not introduced.

The simplified model gathers actual electrical variables I_g , V_g and the AC voltage E_g . In quasi static operation, it is possible to calculate the active power with the classical formula:

$$p_{mes} = P_{max} \sin(\delta) \quad (8)$$

with,

$$P_{max} = \frac{E_g V_g}{X_c + X_g} \quad (9)$$

δ denotes the angle between the VSC voltage E_g and the grid AC voltage v_g .

Based on the simplify model, Fig. 6 depicts the synchronization process of the grid-forming converter subject to a three-phase bolted fault. For a given p^* , there are two equilibrium points (a) and (a'), where the power p_{mes} is equal to its reference p^* . However, only the equilibrium point (a) is stable in small-signal analysis [20]. The initial angle δ_0 is given by,

$$\delta_0 = \arcsin\left(\frac{p^*}{P_{max}}\right) \quad (10)$$

The maximum angle δ_{max} of the operating point a' is expressed as,

$$\delta_{max} = \pi - \delta_0 \quad (11)$$

When the fault occurs, p_{mes} drops to zero. Since the filtering effect is neglected, the frequency deviates instantaneously,

$$\omega_{VSC} = m_p p^* + \omega_{set} \quad (12)$$

Since ω_{VSC} is higher than ω_g (Hyper-synchronism), the angle

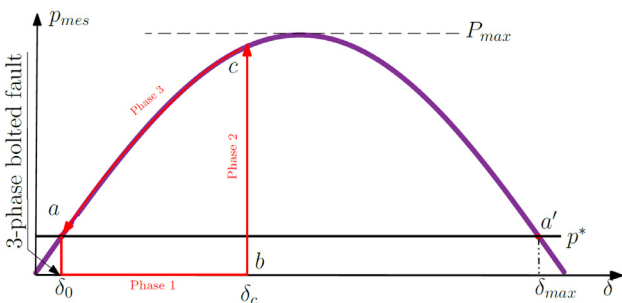
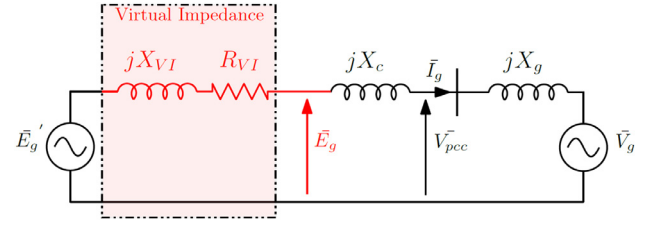
Fig. 6. $p_{mes}(\delta)$ diagram of a droop control based VSC without inertial effect.

Fig. 7. Simplified quasi-static grid-forming model including the VI.

increases (phase 1). When the fault is cleared, the voltage is supposed to recover instantaneously. The consequence is a straight vertical line from the operating point (b) to (c) (phase 2). During this phase $\delta = \delta_c$ (δ_c is called the clearing angle). At the end of phase 2, p_{mes} is higher than p^* leading to $\omega_{VSC} < \omega_g$ (Hypo-synchronism). It results in a decrease of the angle from (c) to the equilibrium point (a) (phase 3), where $\omega_{VSC} = \omega_g$ and $p_{mes} = p^*$.

The stability limit is reached when the clearing angle reaches δ_{max} . This specific angle is defined as the critical clearing angle δ_{cc} [16].

4.2. Transient stability analysis of a VSC based on droop control and virtual impedance

The effect of the virtual impedance is introduced to embed the current limitation in this analysis. Therefore, Fig. 5 is modified when the current exceeds its rated current I_n (Fig. 7).

From Fig. 7, the new expression of the delivered active power including the virtual impedance is expressed as follow:

$$p_{mes} = \frac{E'_g V_g}{\sqrt{X_T^2 + R_T^2}} \sin\left(\arctan\left(\frac{R_T}{X_T}\right) + \delta\right) - \frac{V_g^2 R_T}{X_T^2 + R_T^2} \quad (13)$$

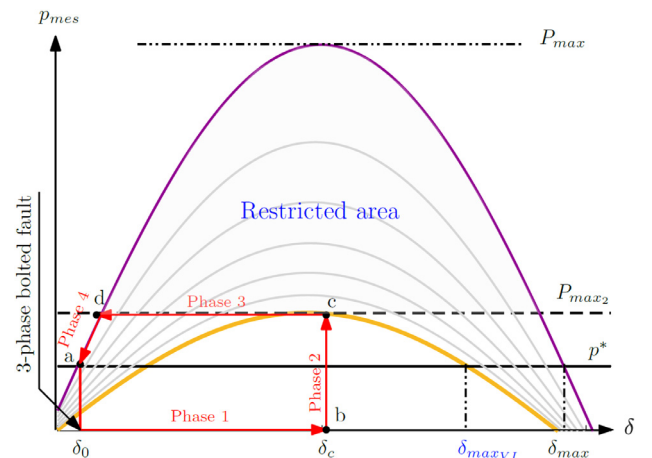
with $X_T = X_c + X_g + X_{VI_{max}}$ and $R_T = R_c + R_g + R_{VI_{max}}$.

$X_{VI_{max}}$ and $R_{VI_{max}}$ are the virtual inductance and resistance defined from the current limit I_{max} . Note that $R_{VI_{max}}$ could be neglected since $X_{VI} \gg R_{VI}$. However, it is considered to have more accurate theoretical results.

The maximum delivered active power including the virtual impedance is given by:

$$P_{max2} = \frac{E'_g V_g}{\sqrt{X_T^2 + R_T^2}} - \frac{V_g^2 R_T}{X_T^2 + R_T^2} \quad (14)$$

Several $p_{mes}(\delta)$ may be drawn on Fig. 8 for different values of the virtual impedance comprised between $Z_{VI_{max}}$ and 0. The sequence of operations is now modified:

Fig. 8. $p_{mes}(\delta)$ diagram of a droop control based-VSC and VI.

- Phase 1 is similar with or without virtual impedance.
- Phase 2, the fault is cleared. As a consequence, the operating point moves first to (c), which is now located on the new curve p_{mes} in (13).
- Phase 3, the operating point moves from (c) to (d). In this phase, Z_{VI} progressively decreases until Z_{VI} is equal to zero (i.e.; when the operating point is equal to (d)).
- Phase 4, the virtual impedance is disabled and the operating point moves from (d) to the equilibrium point (a).

The activation of the virtual impedance results in a decrease of P_{max} to P_{max2} since the total impedance increases. As a consequence, the system remains stable if the fault is cleared before δ_{maxVI} expressed in (15).

$$\delta_{maxVI} = 2\left(\frac{\pi}{2} - \arctan\left(\frac{R_T}{X_T}\right)\right) - \arcsin\left(\frac{p^* \sqrt{X_T^2 + R_T^2} + V_g^2 R_T}{E_g V_g}\right) + \dots \arctan\left(\frac{R_T}{X_T}\right) \quad (15)$$

Based on (15), a new stability limit is defined,

$$\delta_{cc} = \delta_{maxVI} \quad (16)$$

4.3. Transient stability analysis of a VSC based on droop control and current saturation algorithm

The effect of the CSA is now introduced to embed the current limitation in this analysis.

As the current is synchronized on the d -axis, the expression of the delivered active power when the current exceeds its maximum allowable magnitude I_{maxSAT} is expressed as follow:

$$p_{mes} = P_{max3} \cos(\delta) \quad (17)$$

where,

$$P_{max3} = I_{maxSAT} V_g \quad (18)$$

Fig. 9 depicts the evolution of angle in the current saturated mode. The sequence of operations is now:

- Phase 1 is similar with or without the current saturation algorithms.
- Phase 2, the fault is cleared. As a consequence, the operating point moves to (c), which is now located in the new curve $p_{mes} = P_{max3} \cos(\delta)$.

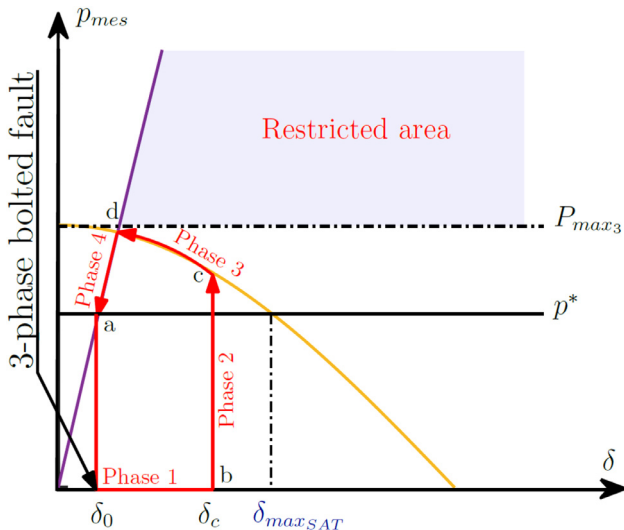


Fig. 9. $p_{mes}(\delta)$ diagram of a droop control based-VSC and CSA.

- Phase 3, the operating point moves from (c) to (d). At this particular operating point, the saturated curve crosses the curve $p_{mes}(\delta)$ presented in Fig. 6. This means that it is possible to recover the normal operating point for the voltage. Then, i_{sdq}^* is equal to i_{sdq}^* , which leads to a natural switching from the saturated mode to the unsaturated one.
- Phase 4, the system moves from (d) to its equilibrium point (a).

The activation of the CSA results in a decrease of P_{max} to P_{max3} . Therefore, the system remains stable if the fault is cleared before δ_{maxSAT} expressed in (19).

$$\delta_{maxSAT} = \arccos\left(\frac{p^*}{P_{max3}}\right) \quad (19)$$

Based on (19), a new stability limit is defined,

$$\delta_{cc} = \delta_{maxSAT} \quad (20)$$

4.4. Critical clearing time calculation

Thanks to the previous analysis and considerations, it is now possible to calculate the critical clearing time t_c , which is the maximum fault duration that guarantees the stability of the system. It is determined based on the critical clearing angle.

Since p_{mes} is equal to zero during the fault, the evolution of the angle in time-domain is given by the following equation,

$$\delta(t) = m_p \omega_b p^* t + \delta_0 \quad (21)$$

The critical clearing time is obtained when δ reaches δ_{cc} , which is equal to δ_{maxVI} when using the VI and δ_{maxSAT} when using the CSA as explained in the previous subsection. It yields:

$$t_{cVI} = \frac{[\delta_{maxVI} - \delta_0]}{m_p \omega_b p^*} \quad (22)$$

$$t_{cSAT} = \frac{[\delta_{maxSAT} - \delta_0]}{m_p \omega_b p^*} \quad (23)$$

To have more general trends, a comparison between the critical clearing times t_{cVI} and t_{cSAT} has been performed in Fig. 10 for large range of operating points.

As affirmed in [7] and proved in this paper, it can be noticed that the virtual impedance ensures a transient stability of the system for larger fault duration compared to the current saturation algorithm (e.g., for $p^* = 0.8$ p.u, the maximum fault duration based on VI is $t_{cVI} = 165$ ms, whereas, the maximum fault duration based on CSA is $t_{cSAT} = 63.7$ ms).

4.5. Validation through time-domain simulations

In this section, the theoretical approach developed previously is verified through time-domain simulations. In Figs. 11 and 12, dynamic $p_{mes}(\delta)$ curves (red and green curves) are drawn for both VI and CSA, respectively.

In Fig. 11 two test cases are performed:

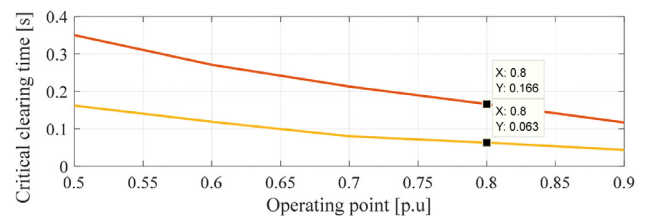


Fig. 10. Comparison between critical clearing time for VI and CSA.

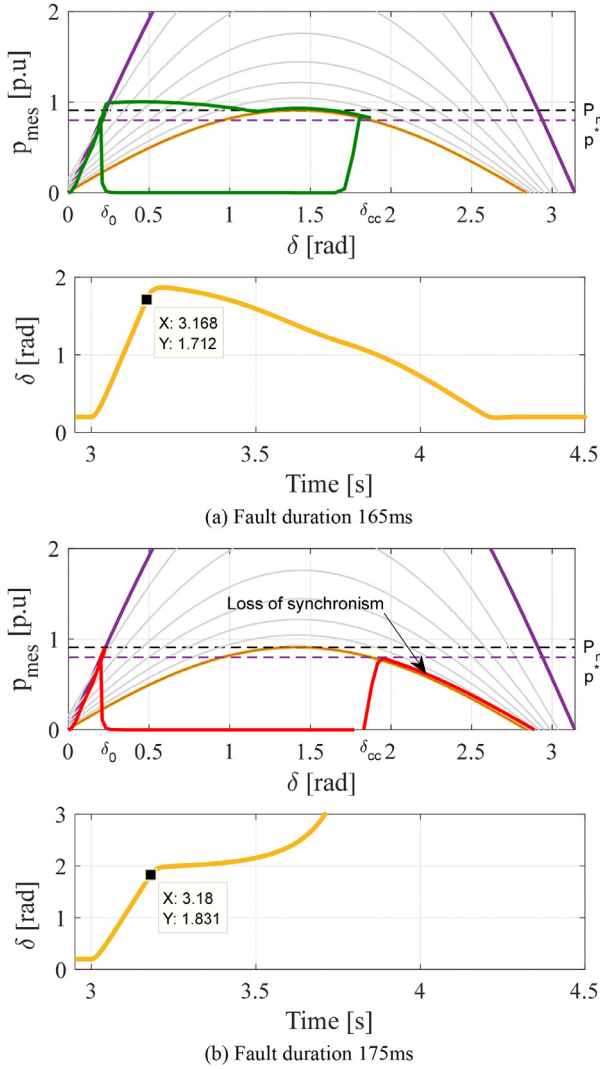


Fig. 11. Dynamic $p_{mes}(\delta)$ curve of the VSC based on droop control and VI.

- The fault duration is equal to the critical clearing time $t_{cVI} = 165$ ms (Fig. 11a).
- The fault duration is higher than the critical clearing time $t_{cVI} = 175$ ms (Fig. 11b).

The first ascertainment from the obtained results confirms the effectiveness of the theoretical approach, where, the system recovers stably to its equilibrium point when $t_{fault} \leq t_{cVI}$ (Fig. 11a), whereas, the system loses the synchronism when $t_{fault} > t_{cVI}$ (Fig. 11b).

Once again, two test cases are performed Fig. 12:

- The fault duration is equal to the critical clearing time $t_{cSAT} = 64$ ms (Fig. 12a).
- The fault duration is higher than the critical clearing time $t_{cSAT} = 74$ ms (Fig. 12b).

Based on the angle evolution, it can be noticed that the system recovers stably to its equilibrium point when $t_{fault} \leq t_{cSAT}$ (Fig. 12a), whereas, the system loses the synchronism when $t_{fault} > t_{cSAT}$ (Fig. 12b).

Some differences can be observed between the theoretical $p_{mes}(\delta)$ curves and the dynamic ones, which are especially linked to the system dynamics i.e., filters, controllers. Nevertheless, the general trend of the evolution is confirmed.

Based on the obtained results, the VI ensures a better transient

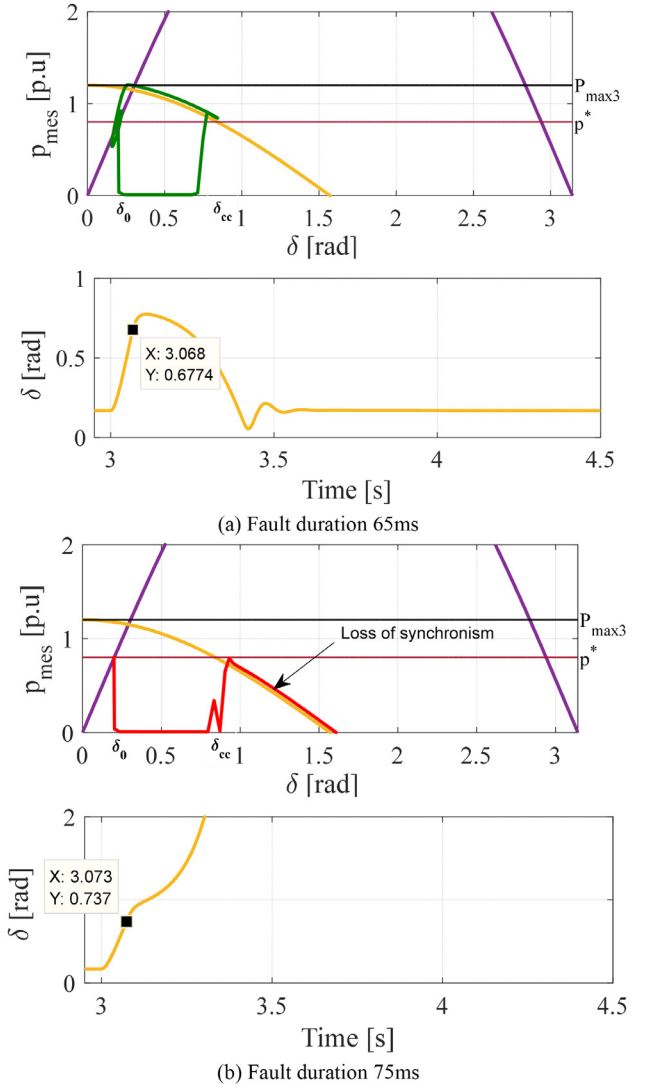


Fig. 12. Dynamic $p_{mes}(\delta)$ curve of the VSC based on droop control and CSA.

stability than the CSA.

5. Hybrid current limiting control and experimental validation

5.1. Hybrid current limiting control principle

It is recalled from the previous sections that:

- CSA is better than VI in terms of current limitation.
- VI is better than CSA in terms of transient stability.

The idea in this section consists in hybridizing both control strategies in order to benefit from their advantages at once.

To avoid the simultaneous operation of the virtual impedance and the current saturation algorithm, the maximum allowable current for the CSA should be higher than the one of VI e.g., $I_{maxSAT} = 1.25$ p.u and $I_{maxVI} = 1.2$ p.u are set as the maximum allowable current for the CSA and VI, respectively.

When a fault occurs, the virtual impedance is activated ($I_s > 1$ p.u). If the current does not exceed I_{maxSAT} , the current saturation remains disabled, otherwise, the current saturation reacts to limit the current to I_{maxSAT} for a few milliseconds waiting for the virtual impedance to decrease I_s until $i_{sdq}^* = i_{sdq}^*$. Once this condition is fulfilled, the current saturation control is disabled naturally without any

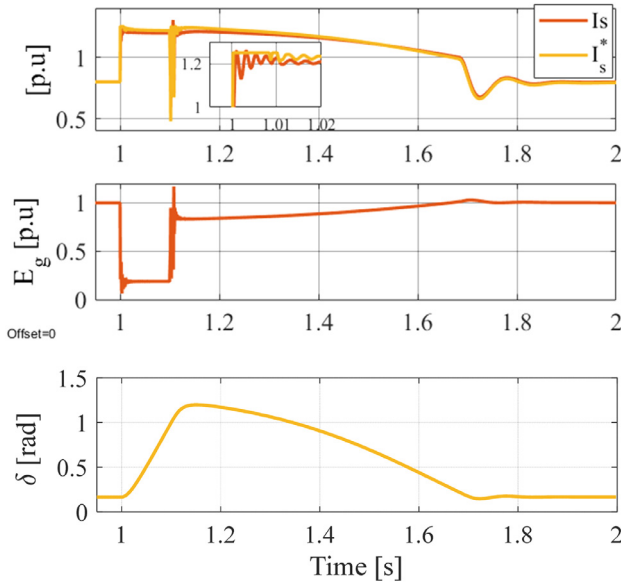


Fig. 13. Three-phase bolted fault at PCC level.

supplementary conditions, and thereby, the current limitation is ensured by the virtual impedance for the rest of the fault duration.

To demonstrate the improvement brought by the proposed HCLC, the same test case in Fig. 4 is performed in Fig. 13.

The results from Fig. 13 show that the CSA limits the VSC output current to $I_{\max\text{SAT}}$ during the first 15ms, then it has been disabled naturally since the virtual impedance decreases the current to $I_{\max\text{VI}}$ for the rest of the fault duration. After the fault clearing, the stable re-synchronization to the AC grid is guaranteed thanks to the VI.

5.2. Experimental validation

The aim of this section is to validate experimentally the theoretical developments. The experimental bench is illustrated in Fig. 14.

The 2-Level VSC (1) is supplied by an ideal 600V DC voltage source (4) and connected to a high bandwidth AC amplifier (5) through an LCL filter (3) as depicted in Fig. 15. The amplifier is used to emulate the AC system (i.e.; 300V ph-ph) as well as to generate the events discussed in this paper (i.e.; a 100% voltage sag emulating a 3-phase bolted fault). The 2-Level VSC is controlled with a dSPACE dS1005 (2) with a 40 μs time step. The switching frequency of the converter is $f_{\text{sw}} = 10\text{kHz}$.

The mockup parameters are listed in Table 2.

The electrical quantities displayed in experimentation are respectively the grid voltages v_g in (abc) frame, the AC voltage amplitude across the capacitor E_g and the VSC output current I_s in d – q frame.

At $t = t_0$, $p^* = 0.7\text{p.u.}$. Then, a 200ms fault is applied. Fig. 16 shows the system dynamics based on the droop control and the HCLC. When

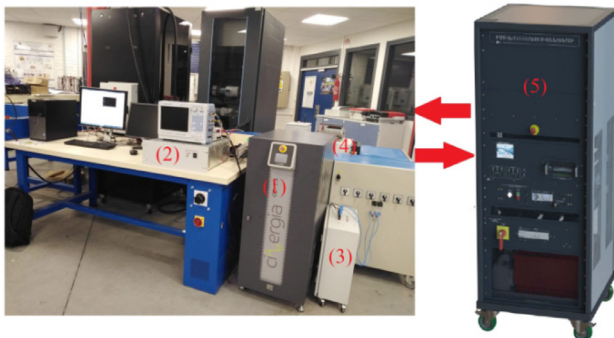


Fig. 14. Mockup presentation.

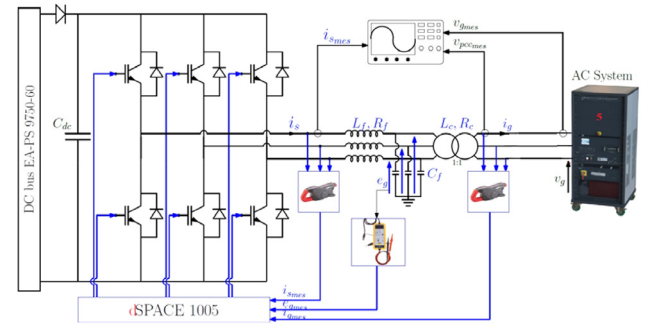


Fig. 15. Functional schema.

Table 2
Mockup parameters.

Symbol	Value
S_n	5.625 kVA
I_n	10.8A
f_n	50 Hz
L_f	10.9mH
L_c	12.9mH
C_f	9.2 μF

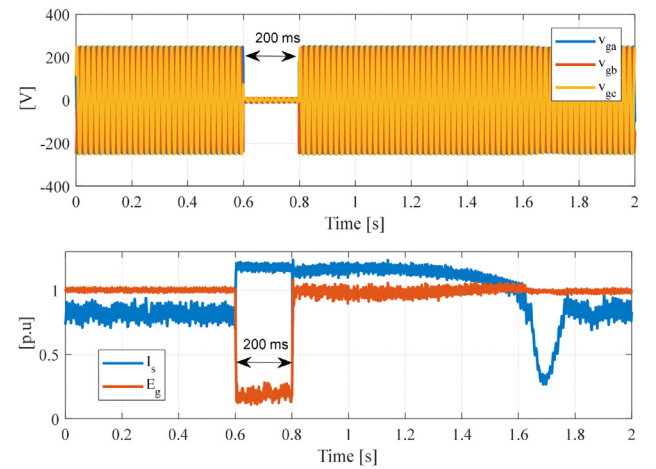


Fig. 16. System dynamics and transient stability of a VSC based on the droop control the HCLC.

the fault occurs, the AC voltage drops, which results in an increase of the VSC output current I_s . This latter is limited to the allowed current magnitude. Once the fault is cleared the system recovers its equilibrium point. The obtained results support the result obtained in Fig. 13.

In the previous test case, the fault duration was less than the critical clearing time. In Fig. 17, 250ms fault duration is applied to the system. During the fault, the current is well limited to the desired value, however, after fault clearing, the system loses the synchronization and becomes unstable. The observed instability phenomenon confirms once again the theoretical approach and the result of Fig. 11.

6. Conclusion

In this paper, the current limitation and the transient stability of a grid forming converter based on droop control are discussed when the system is subjected to large disturbances. Thanks to the analysis and the theoretical approach developed in this paper, the advantages and limitations of the well-known CSA and VI are highlighted. This approach leads to a fundamental conclusion i.e.; the CSA is better for current limitation, whereas, the VI is better in terms of transient stability. Based

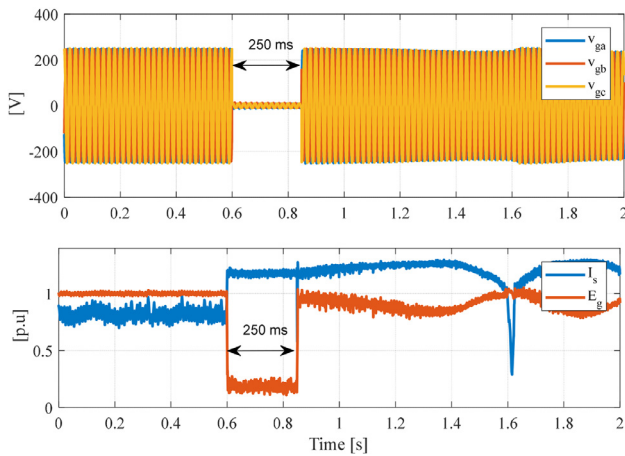


Fig. 17. Transient instability phenomenon of a VSC based on the droop control the HCLC.

on that, hybridization between both strategies is proposed to combine each advantage in one single control strategy called Hybrid Current Limiting Control. Despite that, the transient stability based on VI is better than CSA, the risk of instability remains likely if the fault duration is higher. Therefore, some future works will be devoted to improve the transient stability of grid-forming converters, to develop more theoretical approach for multi-inverter system and to deal with asymmetrical fault. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Competing Interest

None.

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